RAJA

Python Developer & AI/ML Engineer

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Summary

Python Developer and AI/ML Engineer with experience in designing and developing scalable software and intelligent systems. Proficient in backend development using FastAPI, Flask, and Django, as well as in deploying machine learning models and building AI-powered applications using PyTorch, TensorFlow, and Hugging Face. Adept at cloud-based development with AWS, containerization using Docker, and automating deployments with GitHub Actions and Jenkins. Passionate about integrating AI into real-world systems using RESTful APIs, NLP, and LLMs to deliver impactful solutions.

Technical Skills

- Python, Flask, Fast API, Django
- Angular, JavaScript, HTML, CSS
- MySQL, MongoDB, PostgreSQL, ETL
- AWS (S3, EC2, Lambda, SageMaker), GCP
- Machine Learning & AI
- TensorFlow, PyTorch, Hugging Face
- Llama-2, GPT3, GANs, RAG, NLP
- Fine-Tuning, Transfer Learning
- CI/CD (Jenkins, GitHub Actions)
- Kafka (Real-time Data Pipelines, Event-Driven Architecture)
- Git, JIRA, Unit Testing, TDD, BDD

Professional Experience

AI/ML Engineer

Wells Fargo

Charlotte NC — Aug 2023 – Current

Roles & Responsibilities

- Designed and optimized RESTful APIs using FastAPI and Flask for AI-driven financial applications, ensuring low-latency responses and scalable microservices architecture.
- Built scalable Python-based data processing workflows using Pandas, NumPy, and PySpark, automating ETL tasks and feature pipelines for model training and analytics.
- Developed and optimized large-scale ETL pipelines using PySpark and Databricks, processing terabytes of structured and unstructured data for AI-driven analytics.
- Fine-tuned pre-trained LLMs like GPT-3, BERT, and Llama-2 to generate personalized financial recommendations, perform sentiment analysis on financial discussions, and summarize user queries for better decision-making.
- Integrated Generative AI models, including GANs and Retrieval-Augmented Generation (RAG) approaches, to automate content creation, enhance query-answering systems, and improve decision-making processes.
- Trained and deployed large-scale transformer models using PyTorch and Hugging Face Transformers, improving the application's NLP capabilities for intelligent financial guidance.
- Deployed fine-tuned transformer models via Hugging Face's Model Hub, FastAPI ensuring scalable and efficient AI inference.
- Developed responsive, component-based UIs using Angular, integrated with REST APIs for real-time financial data visualization.
- Optimized SQL queries and geospatial database management to support large-scale AI data pipelines, ensuring fast and accurate data retrieval.
- Implemented MLOps practices, including model versioning, monitoring, and CI/CD pipelines for ML deployments in cloud environments.
- Managed containerized AI/ML workflows by designing and deploying Docker containers for reproducible training and inference pipelines
- Monitored AWS EC2 instance performance using CloudWatch, setting up alarms and metrics to track resource utilization and proactively address potential issues.
- Collaborated with cross-functional teams, including data scientists, software engineers, and product managers, to deliver scalable machine learning solutions.
- Participated in Agile workflows, using JIRA for sprint planning and tracking AI project progress while identifying areas for continuous improvement.

Software Engineer
Angel One
Hyderabad, IN Apr 2021 - July 2022
Roles & Responsibilities

- Built RESTful APIs using Flask and Django to integrate machine learning models into web applications, enabling seamless interaction with various data sources, including RDBMS and spreadsheets.
- Developed high-performance RESTful APIs for real-time trading data, ensuring seamless integration with external systems such as market data providers and stock exchanges.
- Utilized NLP techniques to process unstructured financial data, extracting insights from regulatory filings, earnings reports, and market updates.
- Applied machine learning algorithms such as linear regression, multivariate regression, Naive Bayes, Random Forest, K-means, & KNN for data analysis.
- Implemented data deduplication and anomaly detection mechanisms, ensuring high data integrity for trading analytics and compliance reporting.
- Developed and optimized distributed system performance using load balancing, caching strategies, and parallel processing for real-time trading data.
- Developed dynamic and interactive user interfaces using JavaScript, HTML, and CSS to create responsive and engaging web applications.
- Improved database query efficiency in MySQL/MongoDB by optimizing indexing and partitioning strategies.
- Implemented Amazon SQS for efficient message queueing and AWS Lambda for real-time notifications, streamlining the handling of asynchronous trading data and triggering immediate alerts for traders on market events.
- Assisted in the documentation of software components and technical specifications, ensuring comprehensive and understandable documentation for future reference.
- Automated Git workflows using hooks, improving the consistency of commit messages and quality checks during the development process.
- Collaborated in an Agile development environment, actively contributing to code reviews and applying Test-Driven Development (TDD) principles to ensure clean, maintainable, and reliable Python code.

Education

- **Lamar University–** Master's in Computer Science May 2024